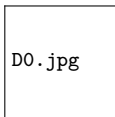


Lecture 05: ER Model: Keys & Weak Entities

Participation Constraints, Role Names, and the Full ER Notation

Z2004: Database Management Systems



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Monday, 9 March 2026

Recap: Lecture 04 — ER Core Concepts

WHERE WE LEFT OFF...

What We Covered

- ▶ **Entities:** real-world objects (rectangles)
- ▶ **Attributes:** properties (ellipses); simple, composite, multi-valued, derived
- ▶ **Relationships:** associations between entities (diamonds)
- ▶ **Cardinality:** Rules of engagement (**1:1**, **1:N**, **M:N**)

Today's Objective

- ▶ **Keys:** How do we uniquely identify a record? (The underline)
- ▶ **Participation:** Is the relationship **mandatory**?
- ▶ **Weak Entities:** When an entity can't stand alone.
- ▶ **ISA Hierarchies:** Handling "Superclasses" and "Subclasses."



Remember the Shape Code? We're adding new rules to these today.

Learning Outcomes

By the end of this lecture, you will be able to:

1. **Keys & Identifiers:** Select Primary Keys; use the underline notation correctly.
2. **Participation Constraints:** Distinguish total vs. partial participation; draw double-line notation.
3. **Relationship Attributes:** Correctly place attributes on relationships (vs. entities).
4. **Weak Entities:** Identify weak entities; draw double-rectangle and double-diamond notation.
5. **Recursive Relationships:** Model unary (self-referencing) relationships with role names.
6. **ISA Hierarchies:** Apply generalization; state disjoint vs. overlapping constraints.

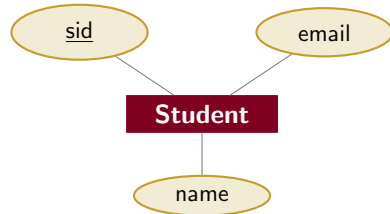
We are still in the **Conceptual Phase**, capturing *what* the data means before writing a single line of SQL.

Primary Keys: The Unique Identifier

For an E-R model to be useful, we must be able to distinguish between individual instances of an entity. This is the role of the **Key Attribute**.

The Rules of Keys

- ▶ **Uniqueness:** No two instances of an entity can share the same value for a key attribute.
- ▶ **Notation:** In Chen's notation, we denote the Primary Key by underlining the attribute name.
- ▶ **Composite Keys:** If a key consists of multiple attributes, underline all of them.
- ▶ **Candidate Keys:** If multiple attributes are unique, we choose one to be the primary identifier for our diagram.

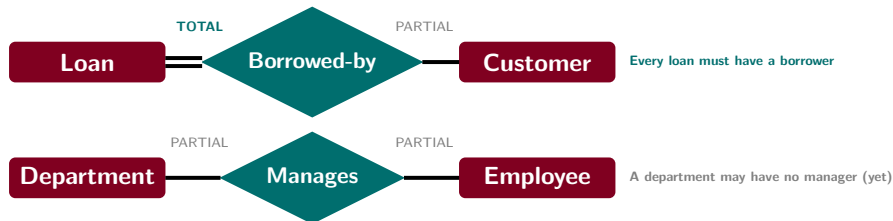


Note: While email is unique, we chose sid as our Primary Key.

This "Underline Rule" carries directly over into the **Relational Schema** where the primary key serves as the

Participation: Total vs. Partial

Participation specifies whether every instance of an entity must participate in at least one instance of a relationship.



Total Participation:

Represented by a **Double Line**. This means the entity *cannot exist* without the relationship.

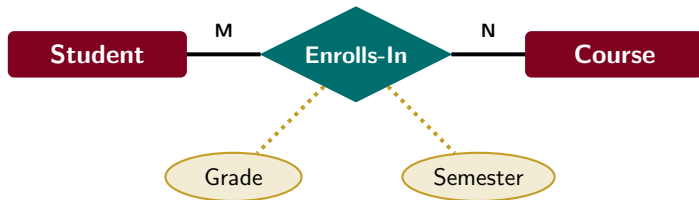
Partial Participation:

Represented by a **Single Line**. This means the relationship is **optional** for the entity.

These lines represent the "Blueprint" logic, determining **NOT NULL** settings in the build phase.

Relationship Attributes

Sometimes an attribute belongs to the *relationship itself*, not to either entity. It only makes sense when the relationship exists.



Rule: Ask “does this describe one entity, or the *combination*?”

A grade only makes sense for a specific (Student, Course) pair.

Because this is an **M:N** relationship, **Enrolls-In** will become its own SQL table. **Grade** and **Semester** will be columns in that new table!

Weak Entities: Identity Without Independence

A **weak entity** cannot be uniquely identified using its own attributes alone. It depends entirely on a **strong (owner) entity**.



Key Notation:

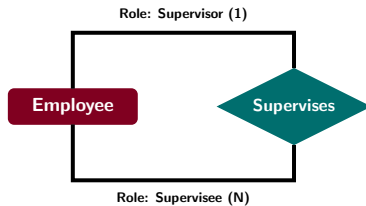
- ▶ **Double rectangle:** Weak Entity
- ▶ **Double diamond:** Identifying Relationship
- ▶ **Dashed underline:** Partial Key (Discriminator)
- ▶ **Double line:** Total participation (Mandatory)

Common Examples:

- ▶ **Room** depends on **Hotel** (Room 101 exists in many hotels, it needs the Hotel ID)
- ▶ **Dependent** depends on **Employee**
- ▶ **Chapter** depends on **Book**

Recursive Relationships & Role Names

A **recursive (unary) relationship** connects an entity type to *itself*. Because the same entity participates twice, we must use **Role Names** to distinguish the two sides.



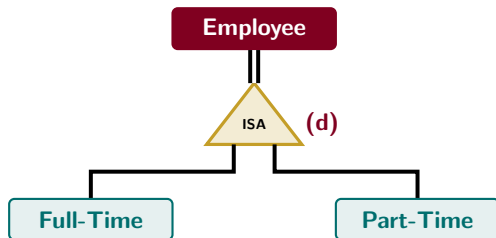
How to read this: One Employee (acting as the *supervisor*) supervises many Employees (acting as *supervisees*).

Other Examples:

Course $\rightarrow$ *is-prerequisite-of* $\rightarrow$ *Course*; *Person* $\rightarrow$ *is-married-to* $\rightarrow$ *Person*

ISA Hierarchies: Specialization

Specialization (Top-Down)



Subclasses **inherit** all attributes of the superclass.

Each subclass can also add its own unique attributes.

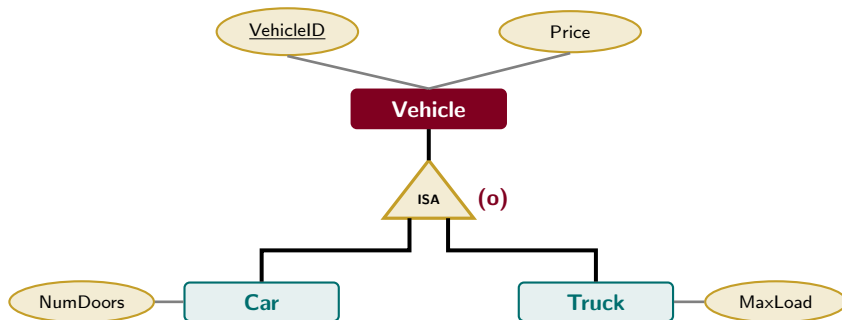
ISA Constraints

- ▶ **Disjoint (d):** An entity can belong to EITHER one subclass OR another, never both (e.g., Full-Time vs. Part-Time).
- ▶ **Overlapping (o):** An entity can belong to multiple subclasses simultaneously (e.g., a Person can be a Student AND an Employee).
- ▶ **Total Coverage:** Every superclass instance *must* belong to a subclass. Shown via a **double line**.
- ▶ **Partial Coverage:** Some superclass instances may not belong to any subclass.

Blueprint to Build: In the relational model, we must decide whether to build one giant table, or three separate tables to represent this hierarchy!

ISA Hierarchies: Generalization (Bottom-Up)

Generalization is the reverse of specialization. We take existing, distinct entities and extract their shared attributes to create a common **Superclass**.



Why Generalize?

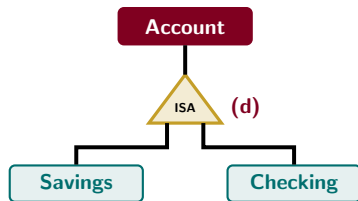
Instead of defining `VehicleID` and `Price` twice, we push them **up** the hierarchy to avoid data redundancy.

Identical Notation:

Whether you design Top-Down or Bottom-Up, the resulting blueprint uses the exact same ISA triangle structure!

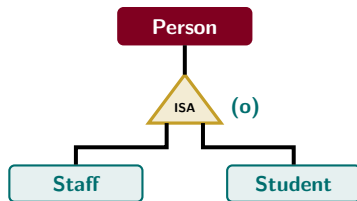
ISA Constraints: Disjoint (d) vs. Overlapping (o)

Disjoint (d)



Rule: An instance can belong to **ONLY ONE** subclass.
Example: A bank account is **EITHER** a Savings account **OR** a Checking account. It cannot be both.

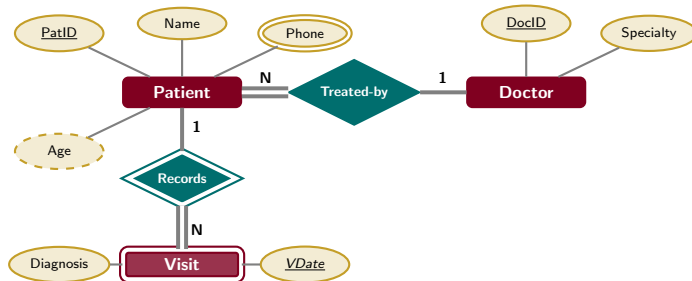
Overlapping (o)



Rule: An instance can belong to **MULTIPLE** subclasses.
Example: A person at a university can be a Staff member **AND** a Student at the exact same time.

Putting It All Together: The Ultimate ER Diagram

Scenario: A hospital tracks Patients and Doctors. Patients *must* have a primary doctor. We also record multiple phone numbers, calculate their age, and track all their hospital Visits (which depend entirely on the Patient existing).



- Strong & Weak Entities
- Identifying Relationship
- Total Participation (Double Lines)
- Cardinality (1:N)
- Primary Keys (Underline)
- Partial Keys (Dashed/Italic)
- Multi-valued (**Double Oval**)
- Derived (**Dashed Oval**)

Class Activity: Design the IITMZ Campus Library

IITMZ Campus Library

The campus library has **Books**, **Students** (Members), and **Librarians**. Students *borrow* books (each loan must record the `DateBorrowed` and `DueDate`). A single book title can have multiple physical **Copies**. A student can borrow multiple books over time. Librarians *manage* the student accounts.

► 1. THINK (2 mins):

List all entities and relationships. Identify at least one **Weak Entity**. Map out the cardinality (1:1, 1:N, M:N) and total/partial participation.

► 2. PAIR (3 mins):

Compare diagrams with your neighbor. Did you both choose the same Weak Entity? What attributes belong on the *Loan* relationship?

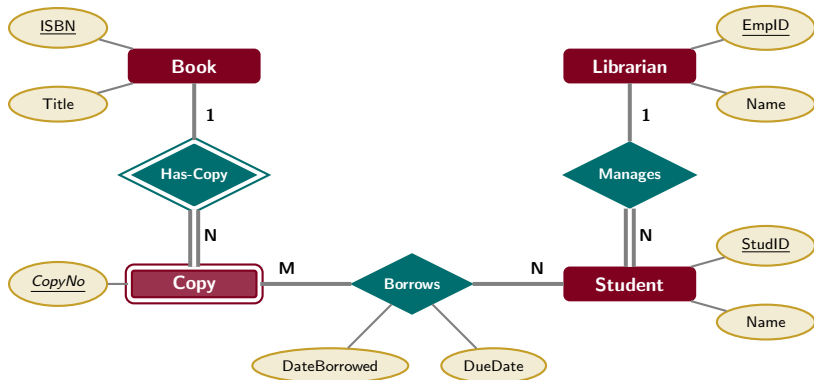
► 3. SHARE (5 mins):

We will draw a combined solution on the board and debate the constraints!

Hint: A specific physical book on the shelf can only be identified by combining the Book's ISBN + a CopyNumber!

Activity Solution: IITMZ Campus Library

Design Notes: Copy is a **Weak Entity** because Copy #1 means nothing without knowing *which* Book it belongs to. The Borrows relationship holds the dates, because those dates apply to the *event* of borrowing, not the student or the book itself.



Common ER Design Mistakes

✗ Putting a relationship attribute on the wrong entity

Grade on a *Student* entity — but a student has different grades in different courses. It must go on the *Enrolls-In* relationship.

✗ Forgetting to double-line a total participation

Every loan must have a borrower — if you draw a single line, you allow “ghost loans” with no customer, which is a business rule violation.

✗ Missing the identifying relationship for a weak entity

A weak entity drawn without a double-diamond identifying relationship is incomplete — the DBMS cannot create the foreign key without it.

✓ Good practice: Read the relationship in both directions.

*“One Doctor **treats** many Patients.”* and *“One Patient is **treated by** exactly one Doctor.”* — gives you the 1:N cardinality.

Concepts Covered

- ▶ **Participation:** Total (double line) vs. Partial (single)
- ▶ **Attributes:** Relationship attributes belong to the pairing/event
- ▶ **Weak Entities:** Double outlines + partial key
- ▶ **Recursive:** Self-joins with explicit role names
- ▶ **ISA Hierarchies:** Generalization, Specialization, (d)/(o) rules

You Can Now

- ▶ Draw a **complete** ER diagram from an English description
- ▶ Correctly classify participation and cardinality constraints
- ▶ Identify and correctly model weak entities
- ▶ Apply ISA notation with appropriate inheritance rules

Lecture 06: Extended ER (EER) Modeling: Aggregation and Advanced ISA.

Assignment 1: Design the university ER diagram in `draw.io` using today's constraints.